Customization of the CVA6 core

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Outline

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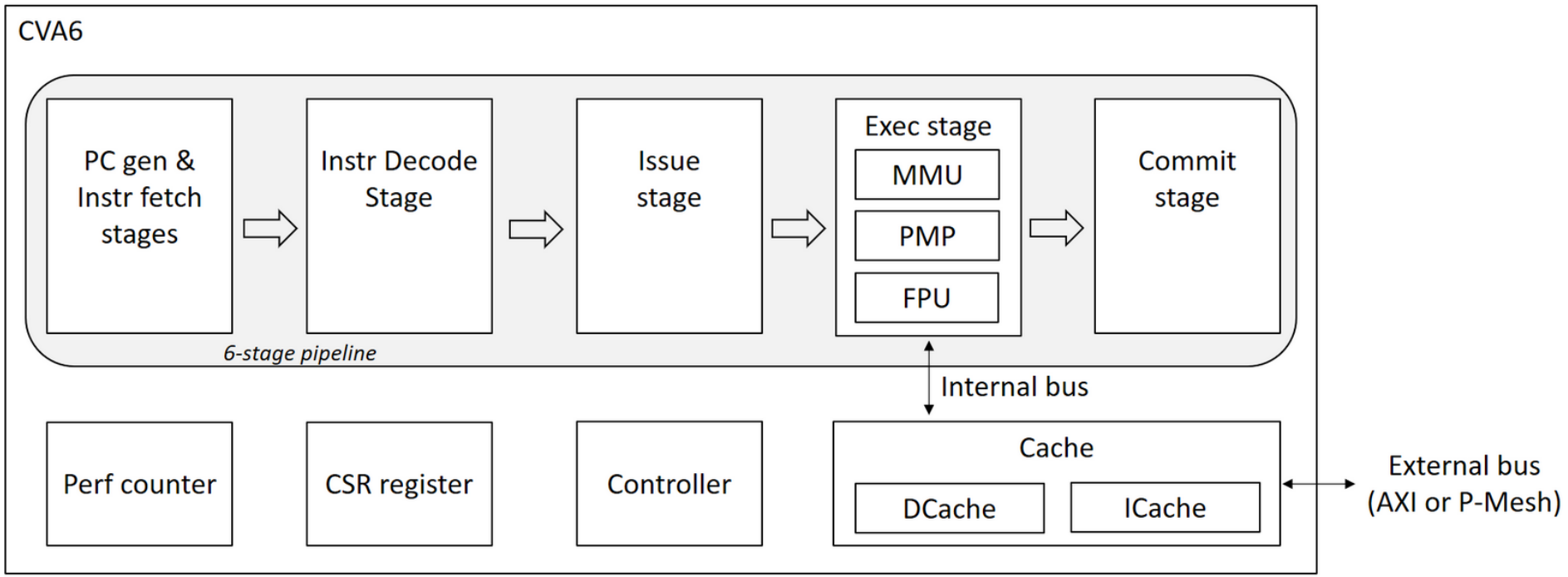
The CVA6 Core

A 6-stage RISC-V Open-Source core

- Industry Standard: CVA6 is a single-issue, in-order design maintained by the OpenHW Foundation (with Thales).
- Versatility: Ideal for ISA extensions (Virtualization), prototyping, and integration into complex multi-core structures like OpenPiton.
- Local Implementation: Internal dual-core prototype developed by the Inria Benagil team for FPGA deployment.

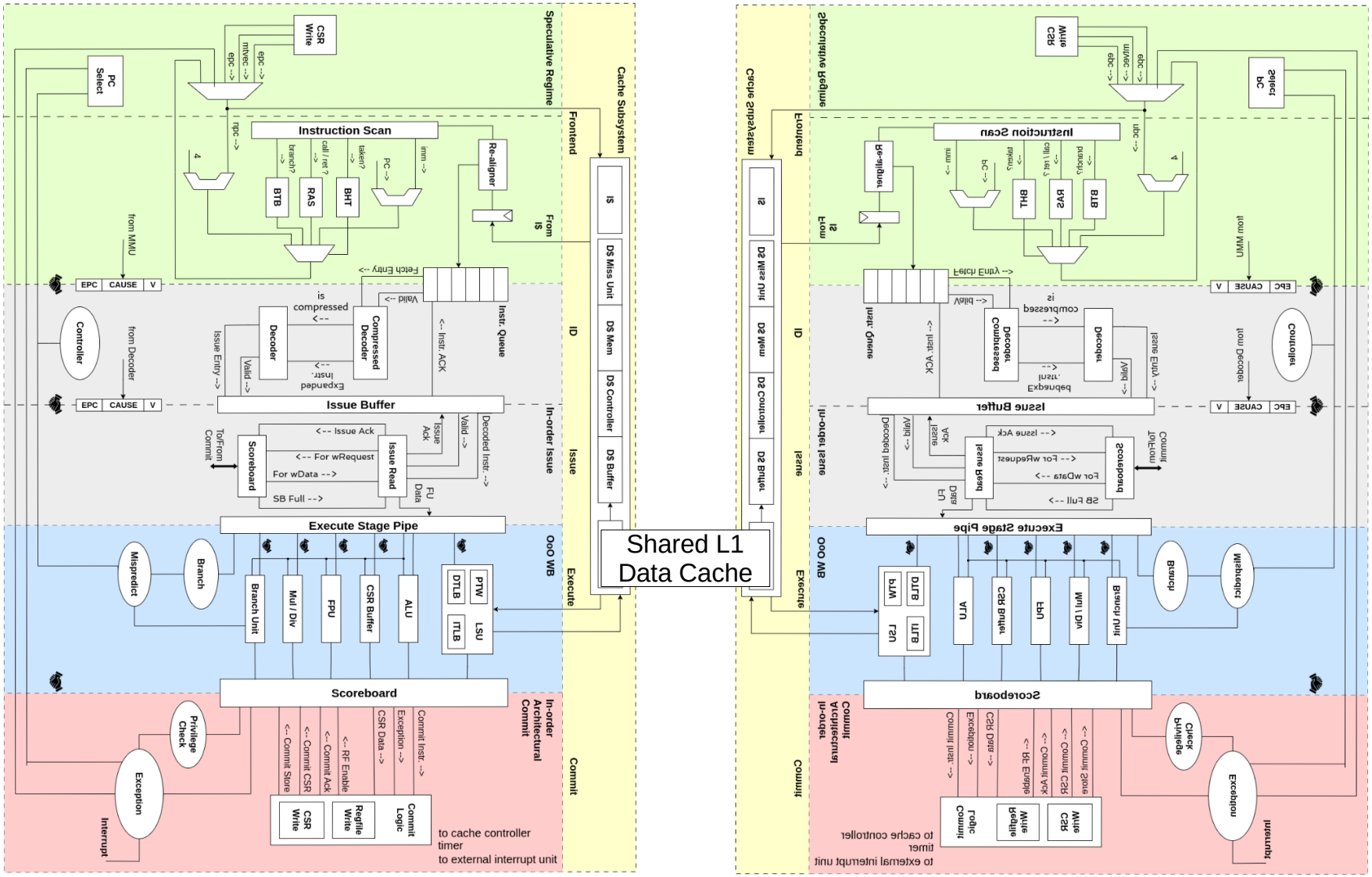
The CVA6 Core

Original Single Core Version



The CVA6 Core

Benagil Team's Dual Core Version



Problem Statement

Problem

The dual-core system is limited by a 1-request-per-cycle bandwidth at the L1D cache.

Research Objectives

- How does the sharing of functional units (starting with the FPU) between two CVA6 cores affect the trade-off between hardware area efficiency and multi-core performance?
- How to measure it accurately.

Project Roadmap

Stage 1 (Baseline Measurements)	Stage 2 (Microarchitectural Study)	Stage 3 (Performance Framework Development)	Stage 4 (Core Modification & Decoupling)	Stage 5 (Arbitration Module Design)	Stage 6 (System Integration & Evaluation)
Deployment of CVA6 on the Pynq Z2	Analysis of Data and Instruction cache sizes	Automated framework for performance measurement	Development of a modified CVA6 version	Development of the shared resource arbitrator	Evaluation of performance and FPGA utilization (FF/LUT/BRAM)
Baseline SoC Utilization measurements (FF/LUT/BRAM)	Study of Branch Predictor and TLB size influence	Integration of NAS Parallel Benchmarks	Implementation of an external module interface [optional]	Implementation of pipeline stalling logic [optional]	Evaluation of performance NPB and Unixbench
Baseline NPB and Unixbench measurements		Integration of UnixBench suite	Enabling connection for FPU management [optional]	Routing of results between multiple cores [optional]	

Outline

How do We Measure Performance?

What do we measure?

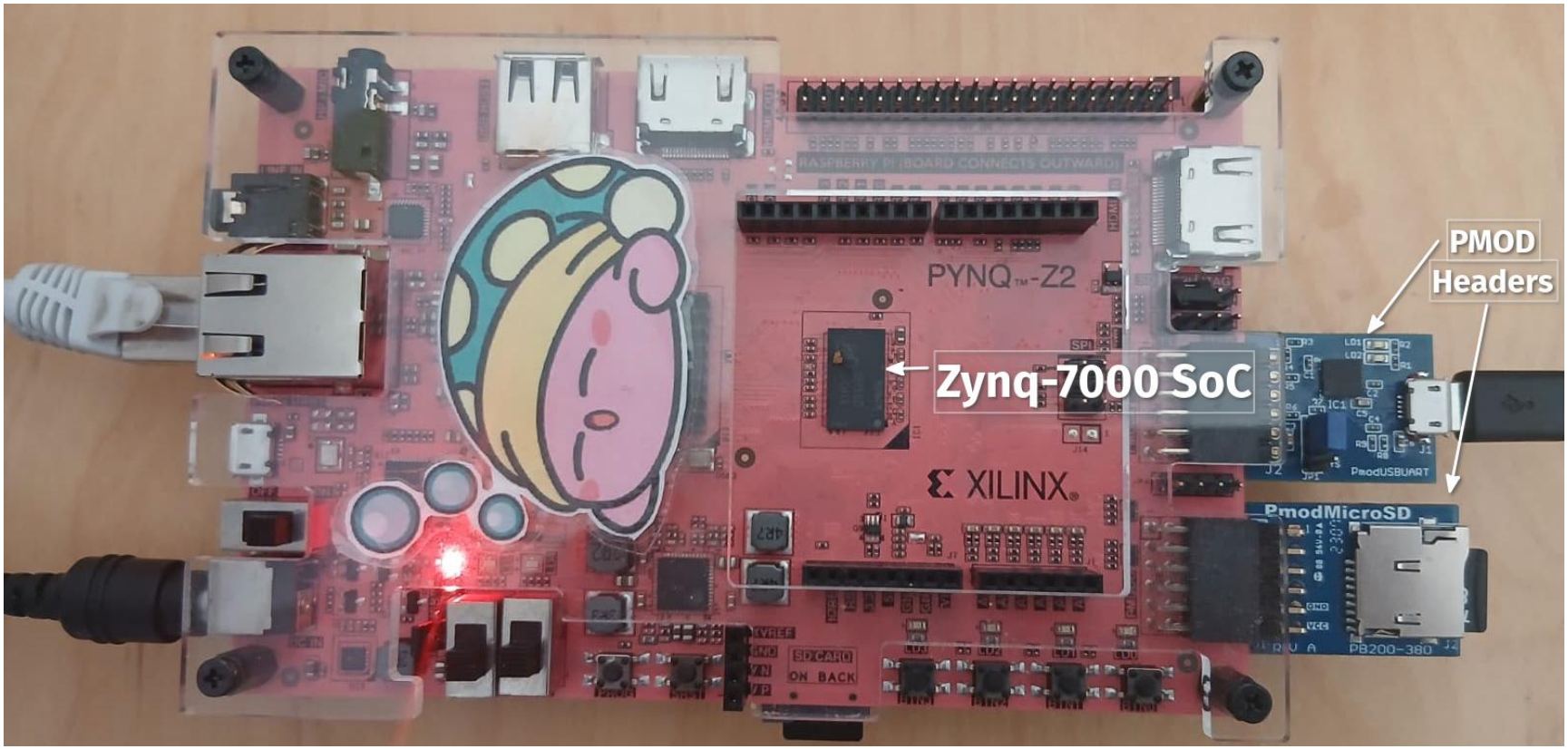
We evaluate performance using well-known workloads: UnixBench and the NAS Parallel Benchmarks (NPB). For each benchmark, we also measure instructions per cycle (*IPC*).

$$IPC = \frac{\text{Number of Instructions}}{\text{Number of cycles}} \tag{1}$$

- Hardware performance counters. *Single Core* available performance counters:
  - Number of Instructions
  - Number of cycles
- Benchmark scores:
  - NPB: Mop/s (Millions of Operations per Second)
  - Unixbench: Loops per second (lps), KB/sec, and operations per second

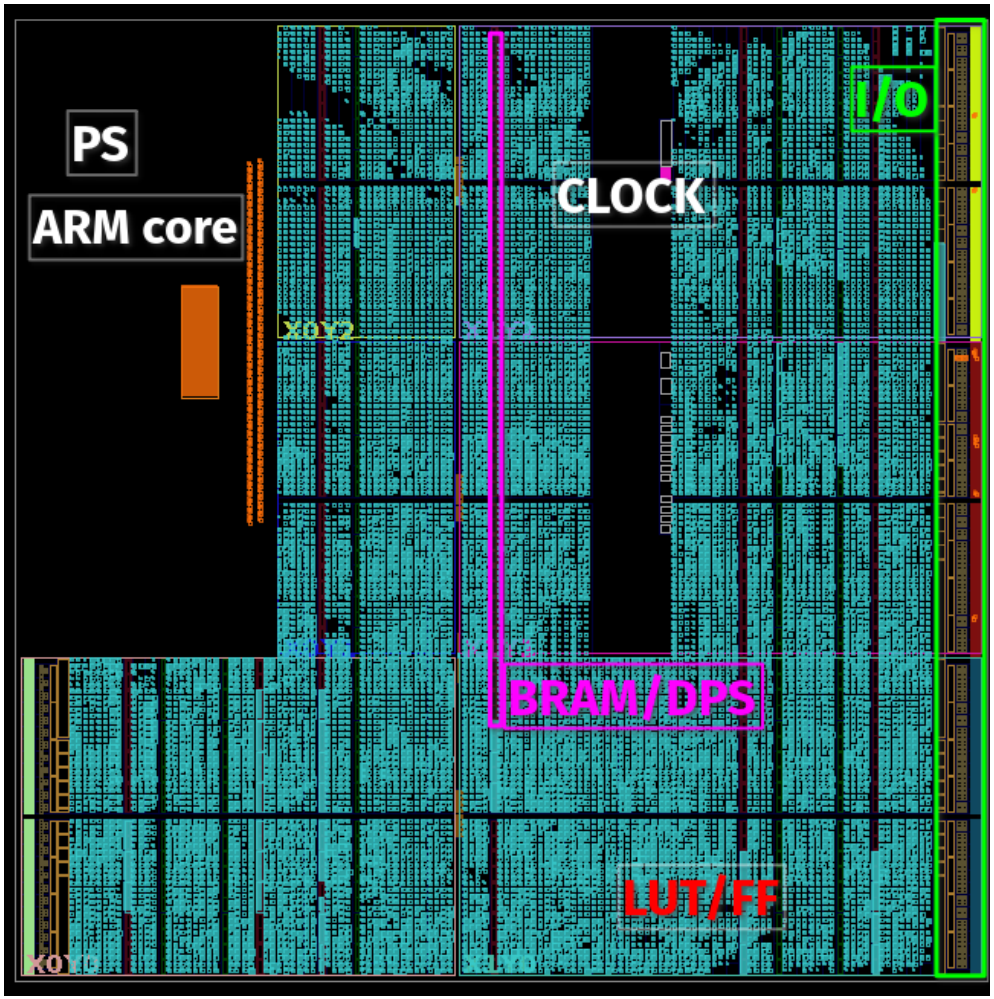
How do We Measure Performance and Resource Usage?

Experimental Setup



CVA6 *Single Core* deployment on a PYNQ-Z2 FPGA with PMod Modules

How do We Measure Resource Usage?

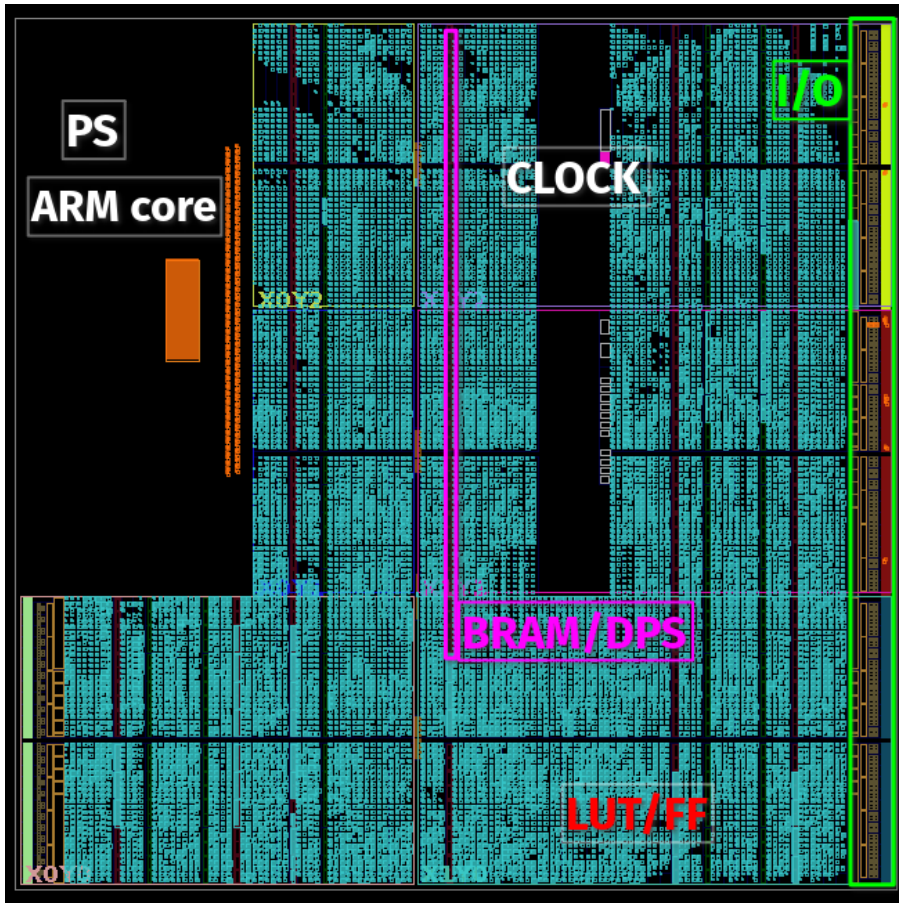


Zynq-7000 SoC

The SoC (System on a chip) is mapped onto the FPGA's programmable logic, utilizing the following hardware resources to implement the CVA6 design:

- Look Up Tables (LUT) for combinatorial logic.
- Flip Flops (FF) for state storage
- Block Random Access Memory (BRAM) for fast on-chip memory
- Digital Signal Processor (DSP) for high-performance math calculations

How do We Measure Resource Usage?

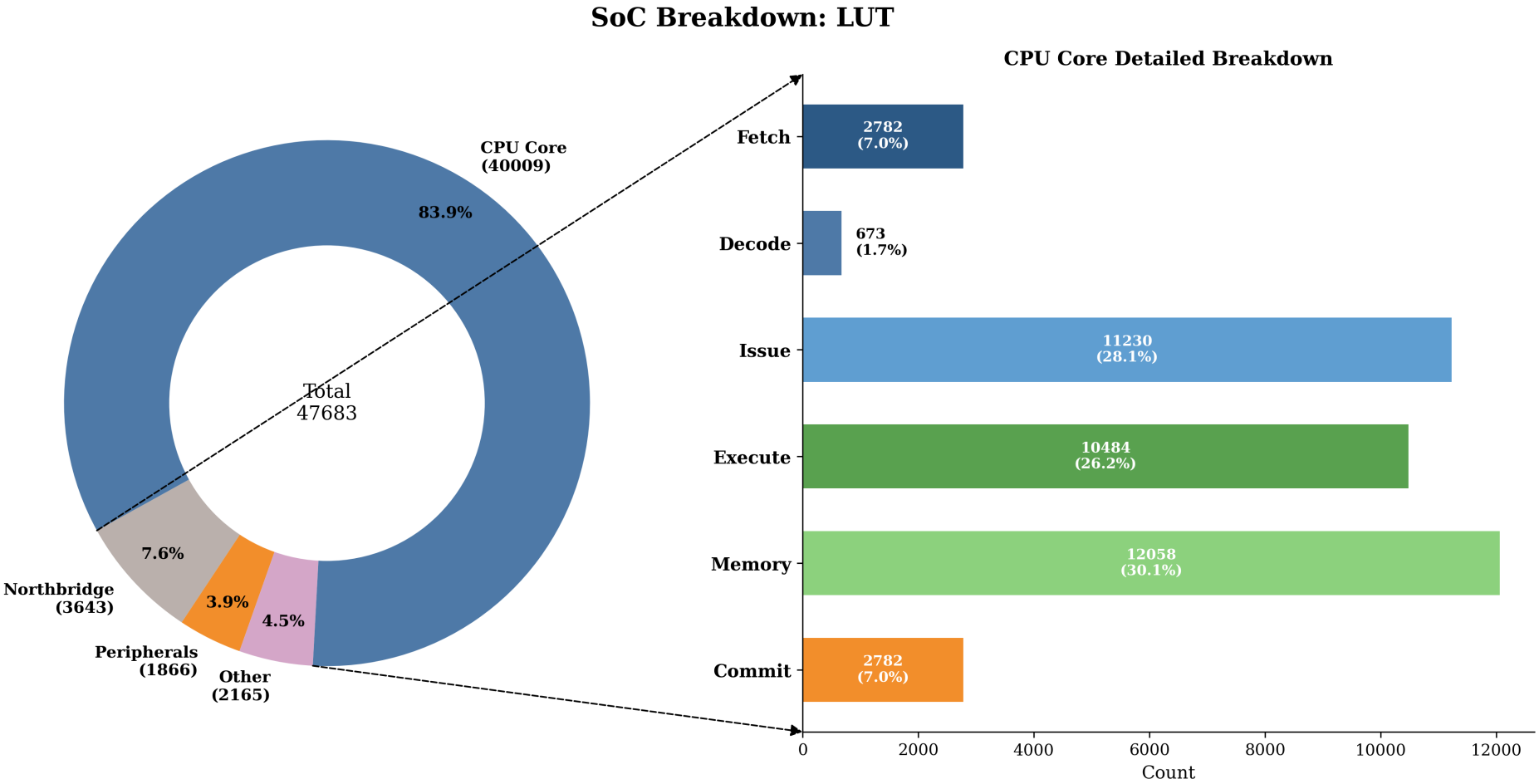


FPGA Area
- LUT
- FF
- DSP
- BRAM

CVA6 stages
- Fetch
- Decode
- Issue
- Execute
- Memory
- Commit

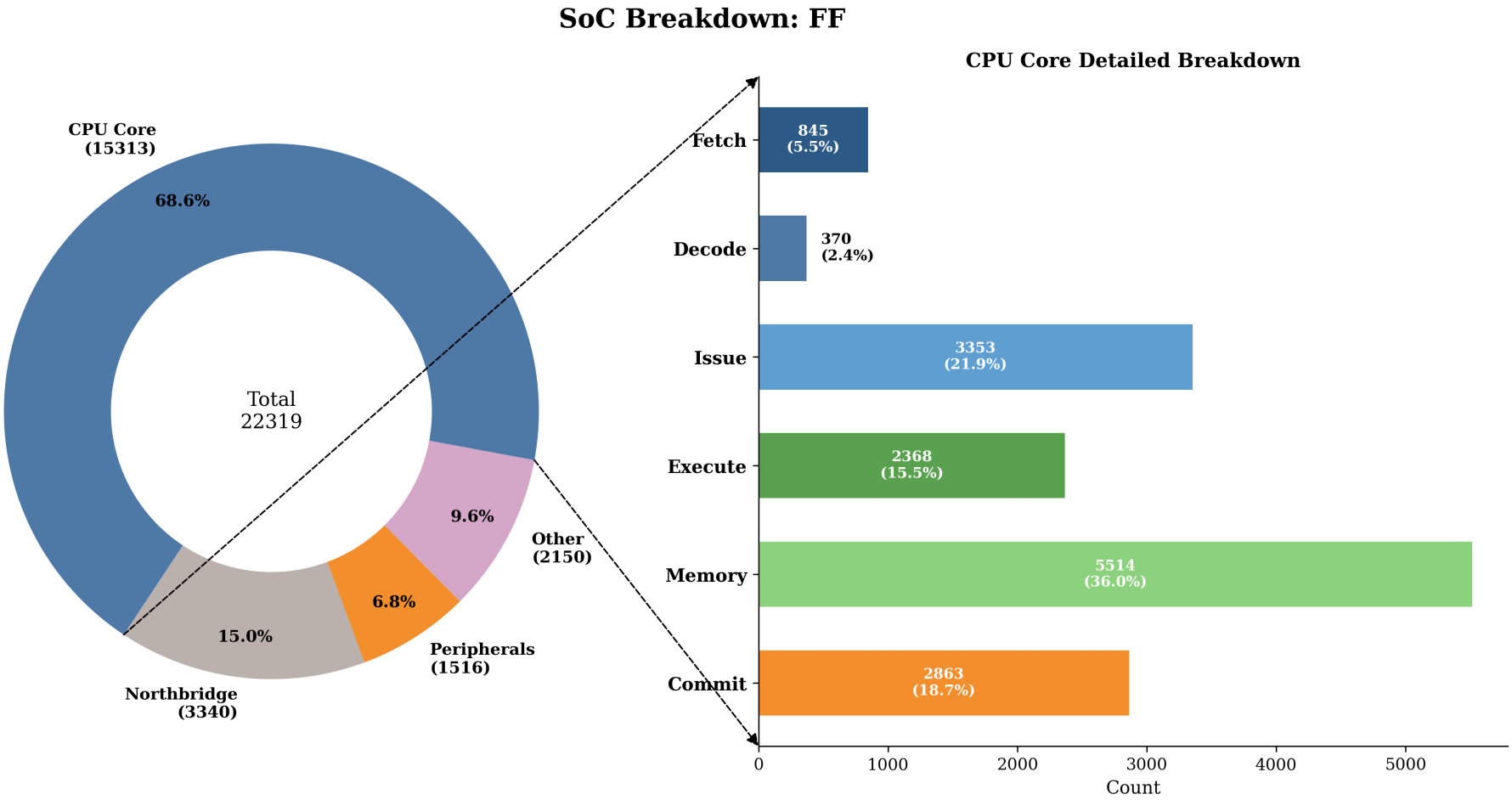
How do We Measure Resource Usage?

Initial LUT Usage Measurement



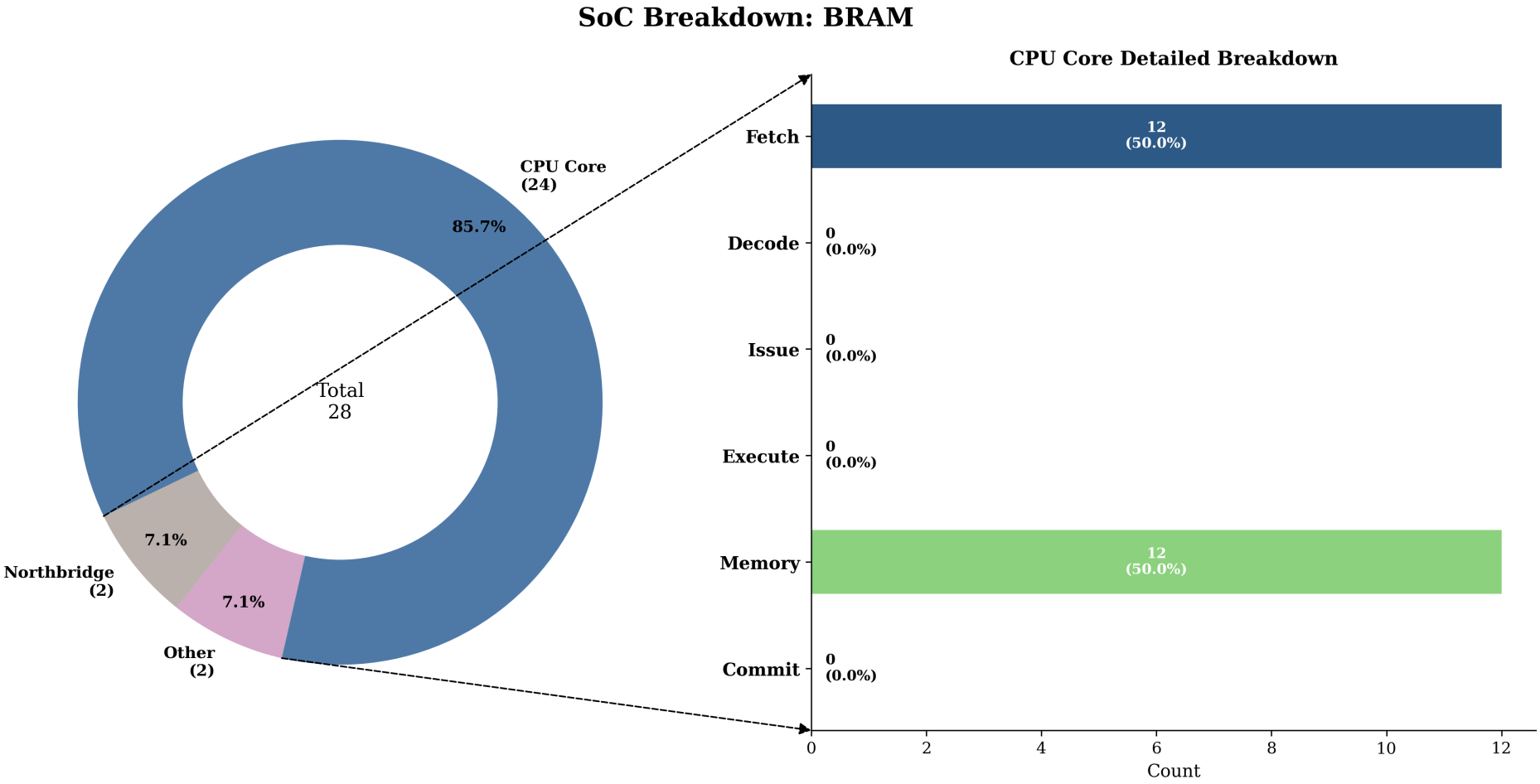
How do We Measure Resource Usage?

Initial FF Usage Measurement



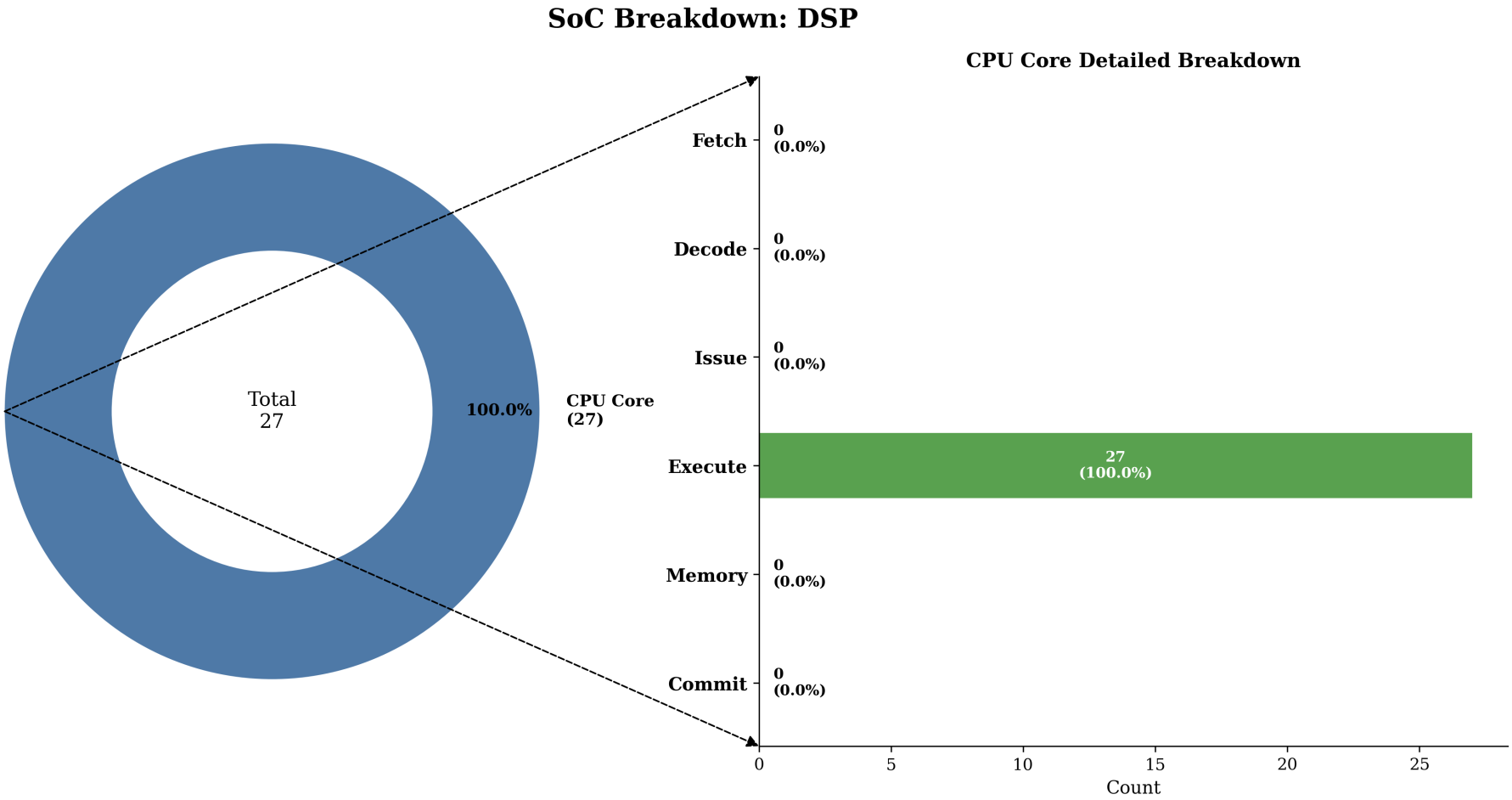
How do We Measure Resource Usage?

Initial BRAM Usage Measurement



How do We Measure Resource Usage?

Initial DSP Usage Measurement



Outline

Baseline Architecture

To study the effects of the micro-architectural parameters the following baseline parameters were used.

Parameter	Size	CVA6 Stage
Data Cache	$4096[bytes]$	Memory
Data Cache Associativity	4	Memory
Data Cache Line Width	128	Memory
Instruction Cache	$4096[bytes]$	Fetch
Instruction Cache Associativity	4	Fetch
Instruction Cache Line Width	128	Fetch
Branch Target Buffer (BTB) Entries	16	Fetch
Branch History Table (BHT) Entries	16	Fetch
Retirement Width of the CPU	2	Commit

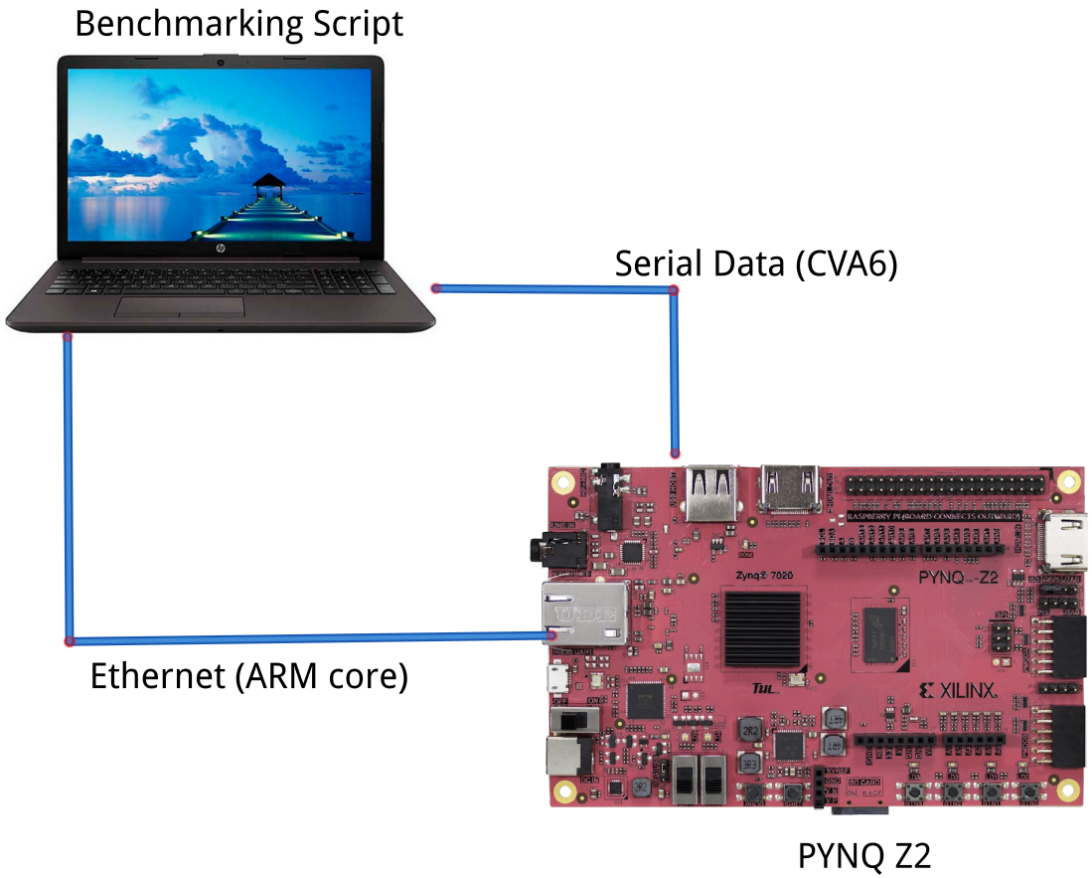
Parameter Sweeps

And the following parameters sweeps were proposed to study their effect on performance and resource utilization.

Sweep	Range	CVA6 Stage
Data Cache	$128[bytes] \rightarrow 256k[bytes]$	Memory
Data / Instruction Cache Associativity	$1 \rightarrow 16$	Memory / Fetch
Instruction Cache	$2k[bytes] \rightarrow 256k[bytes]$	Fetch
BTB Entries	$8 \rightarrow 32k$	Fetch
BHT Entries	$16 \rightarrow 16k$	Fetch

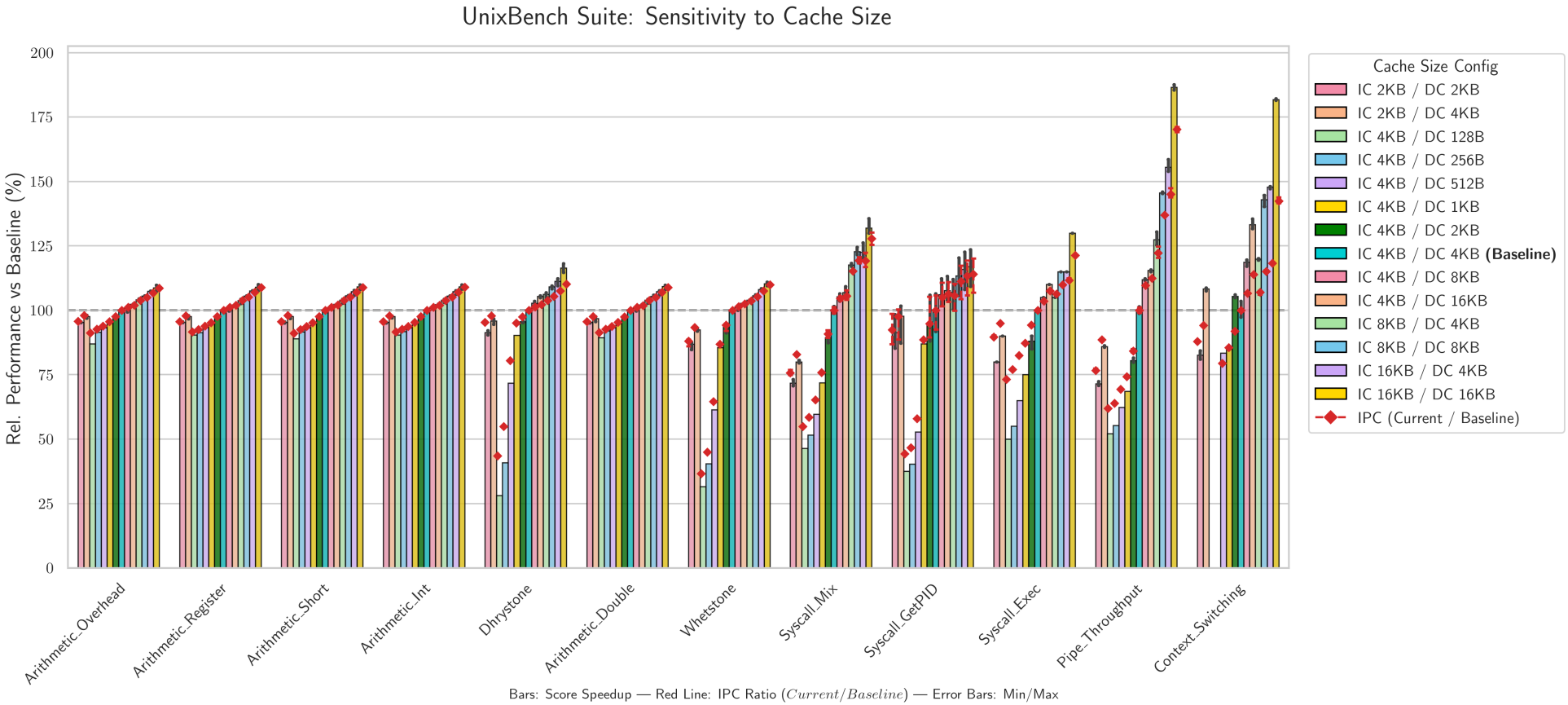
Automated Benchmarking Framework

To automate design space exploration, we developed a framework which uses the ARM core and the CVA6 core, the full run of the benchmarks for all the parameter sweeps takes 21 hours.

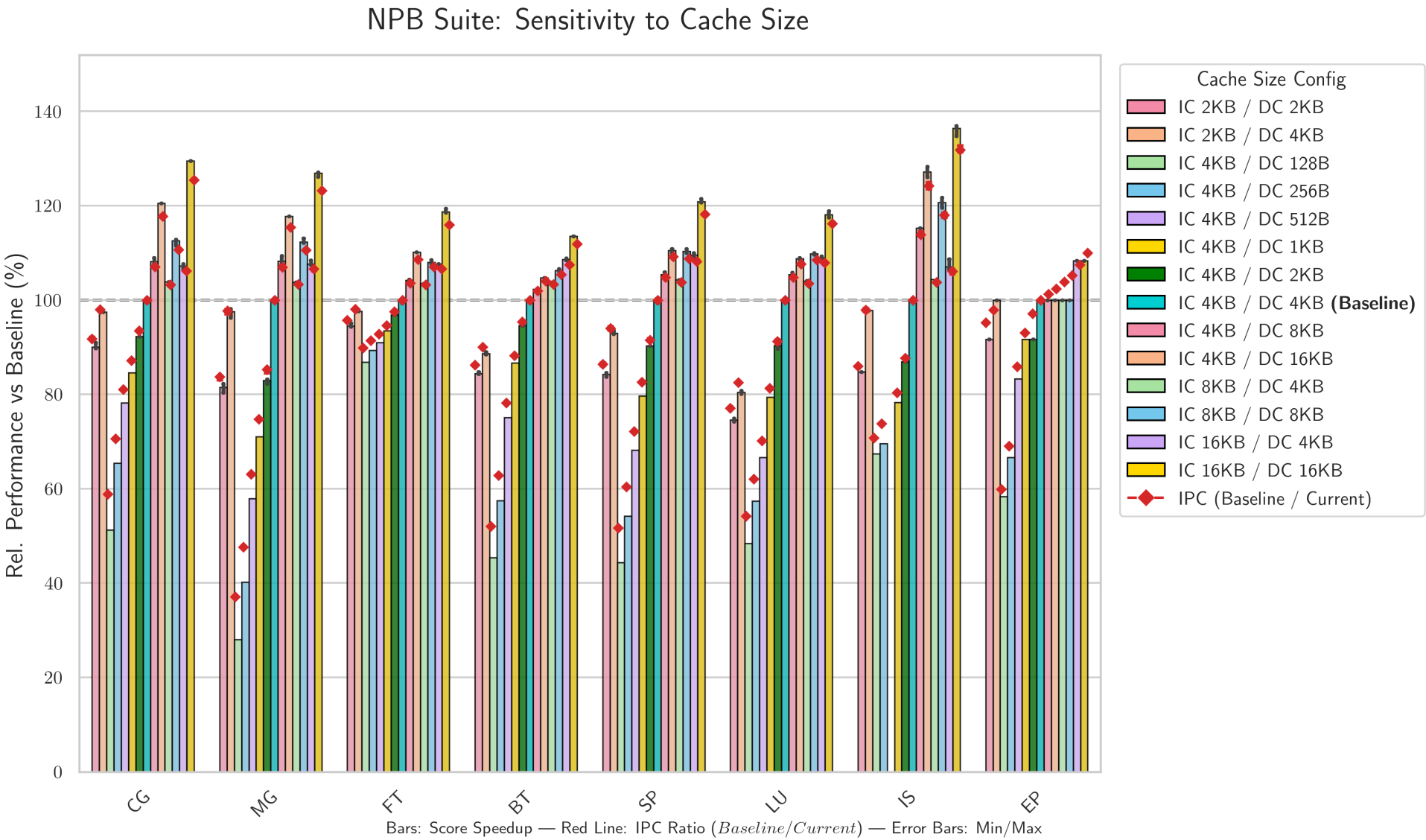


Outline

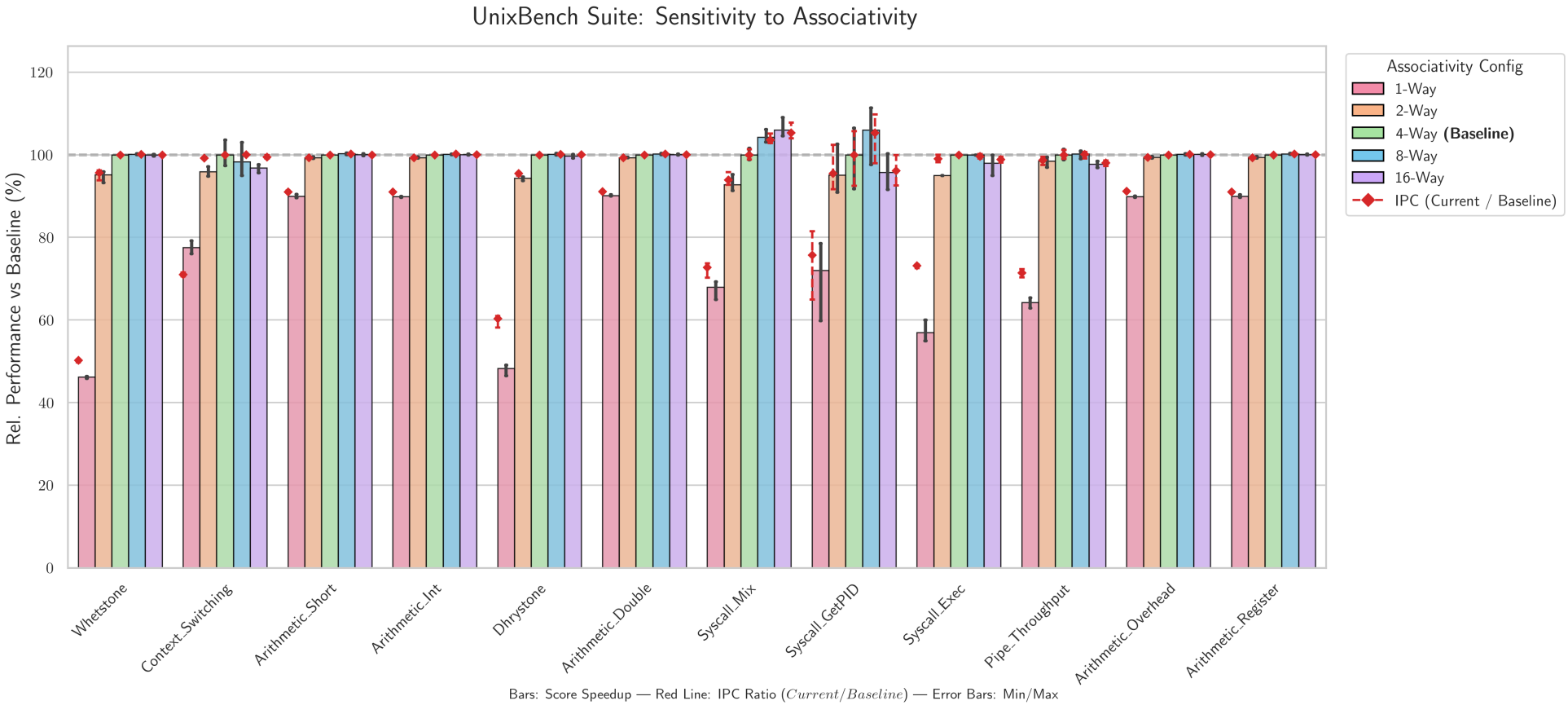
Performance Dynamics



Performance Dynamics



Performance Dynamics



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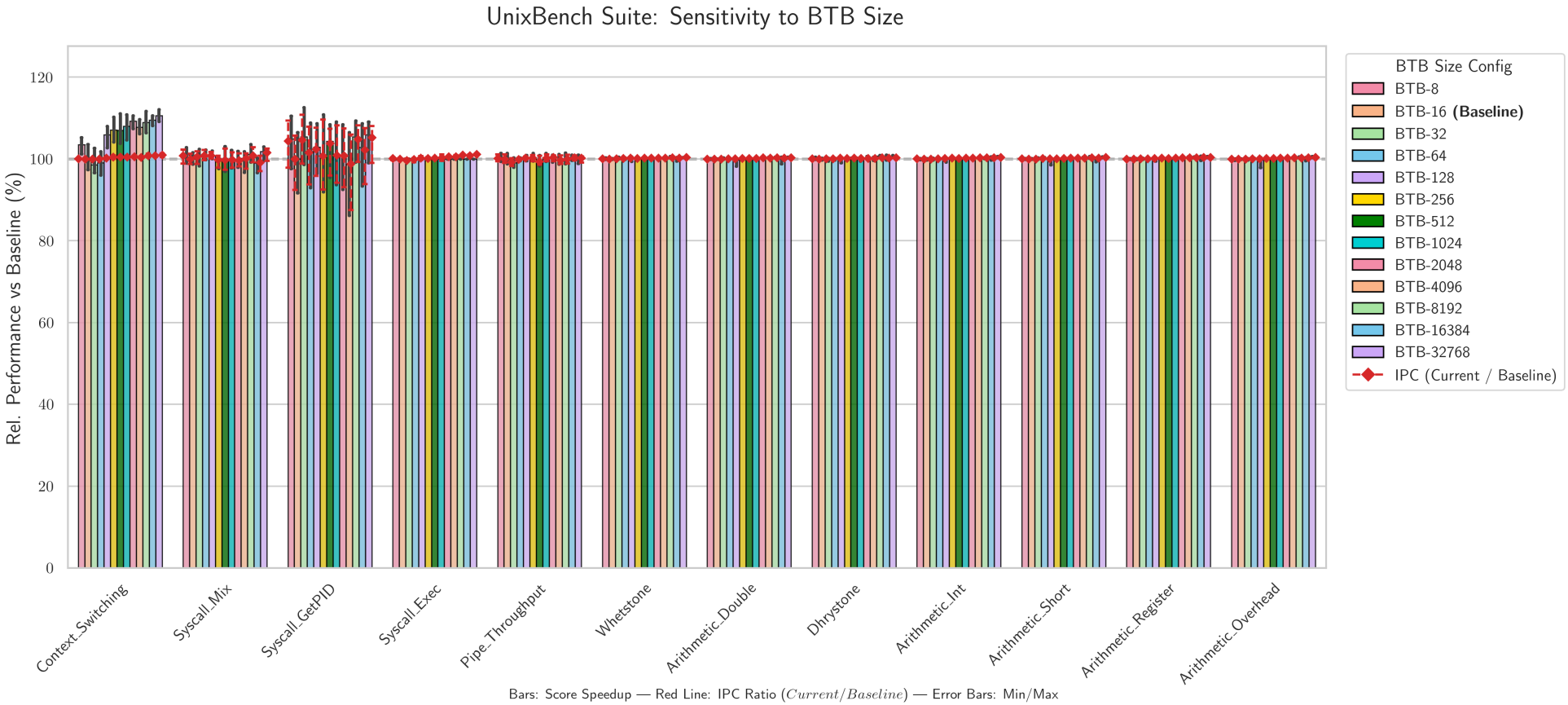
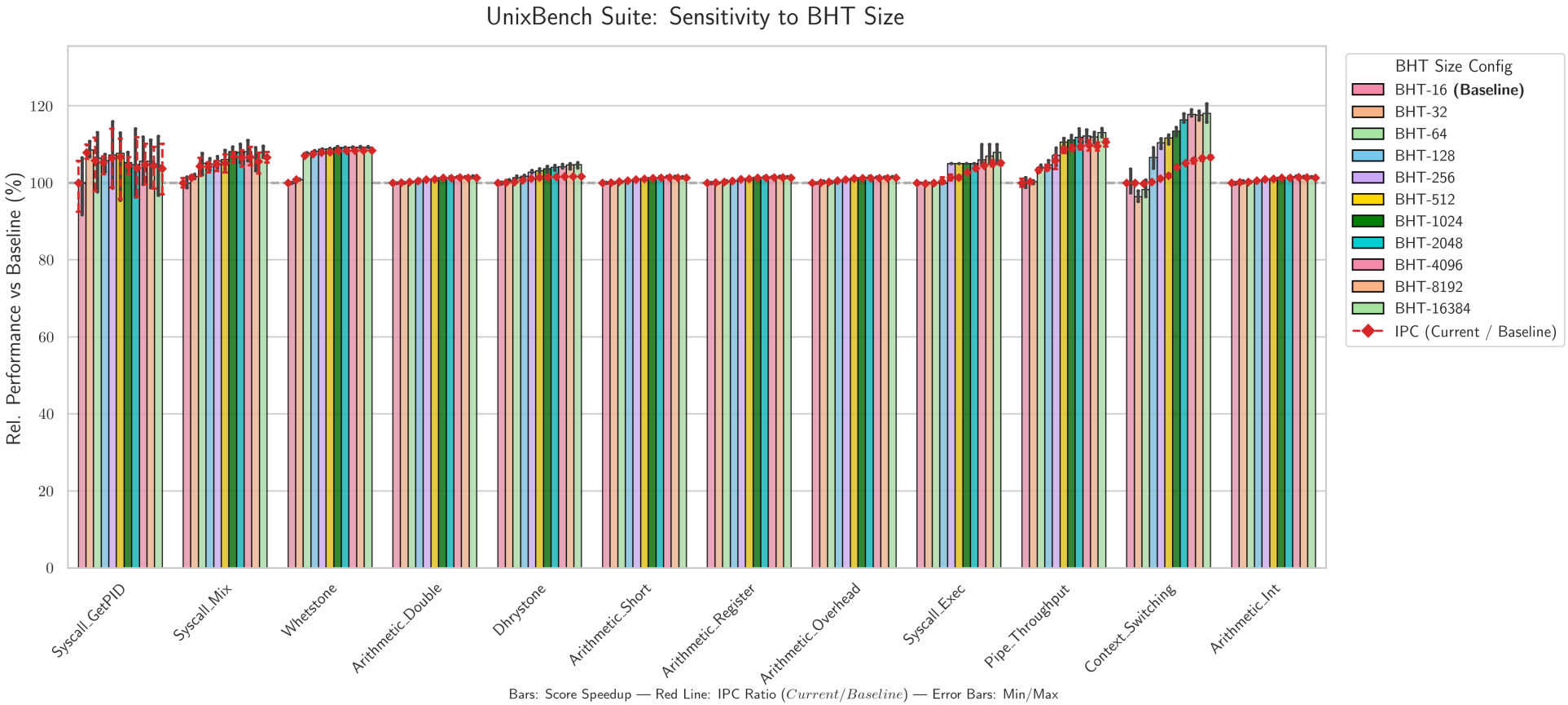
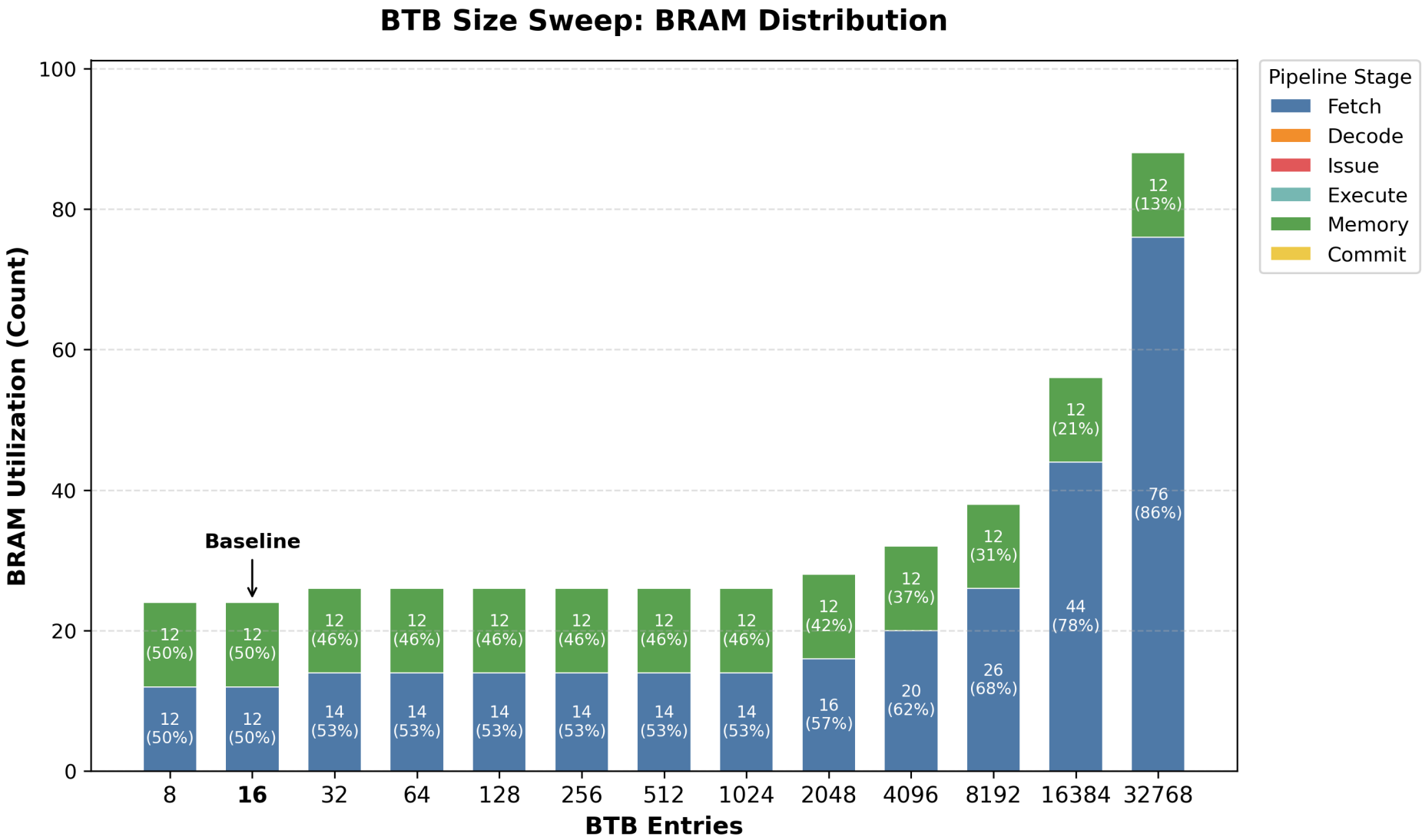


Performance Dynamics



Resource Usage Dynamics



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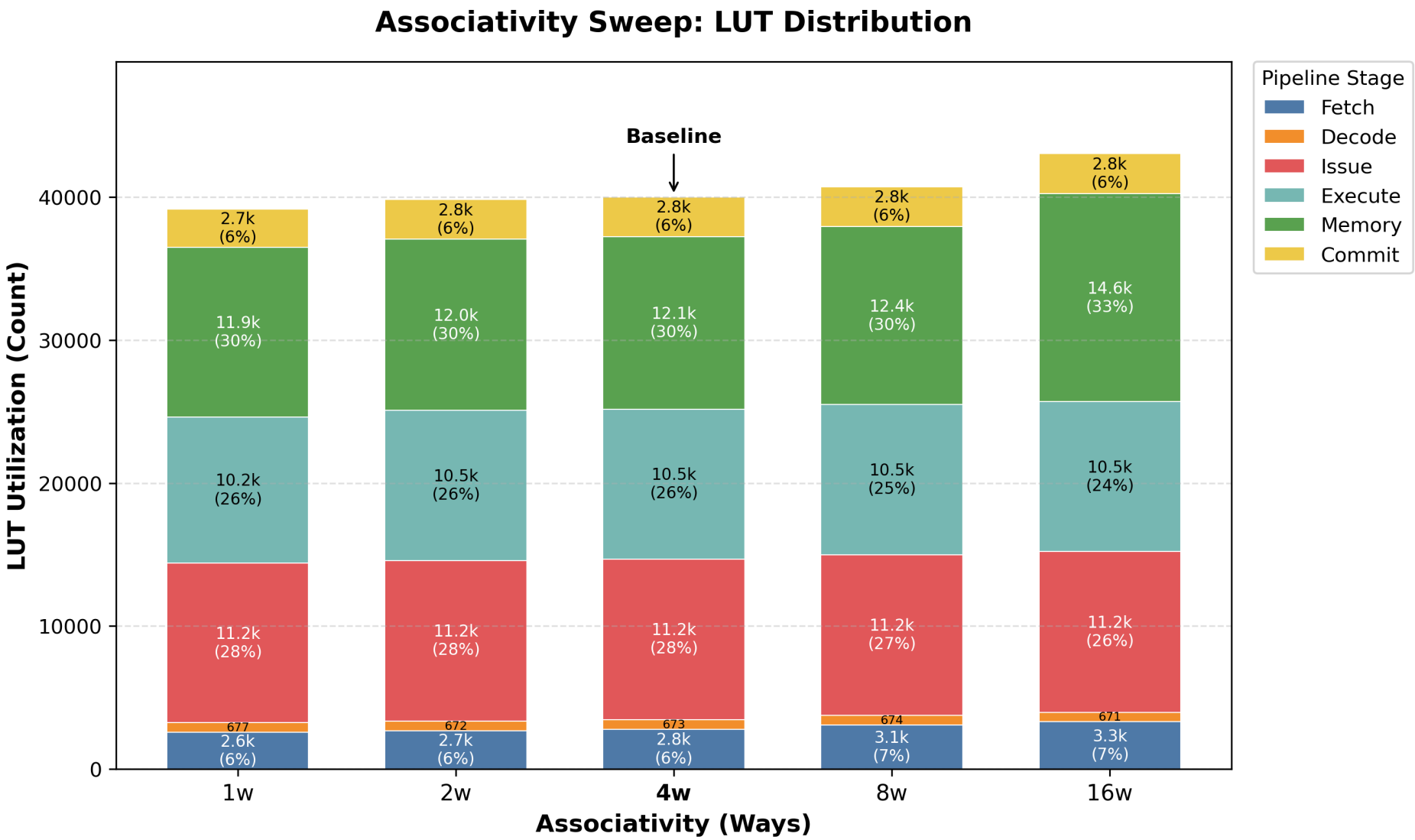
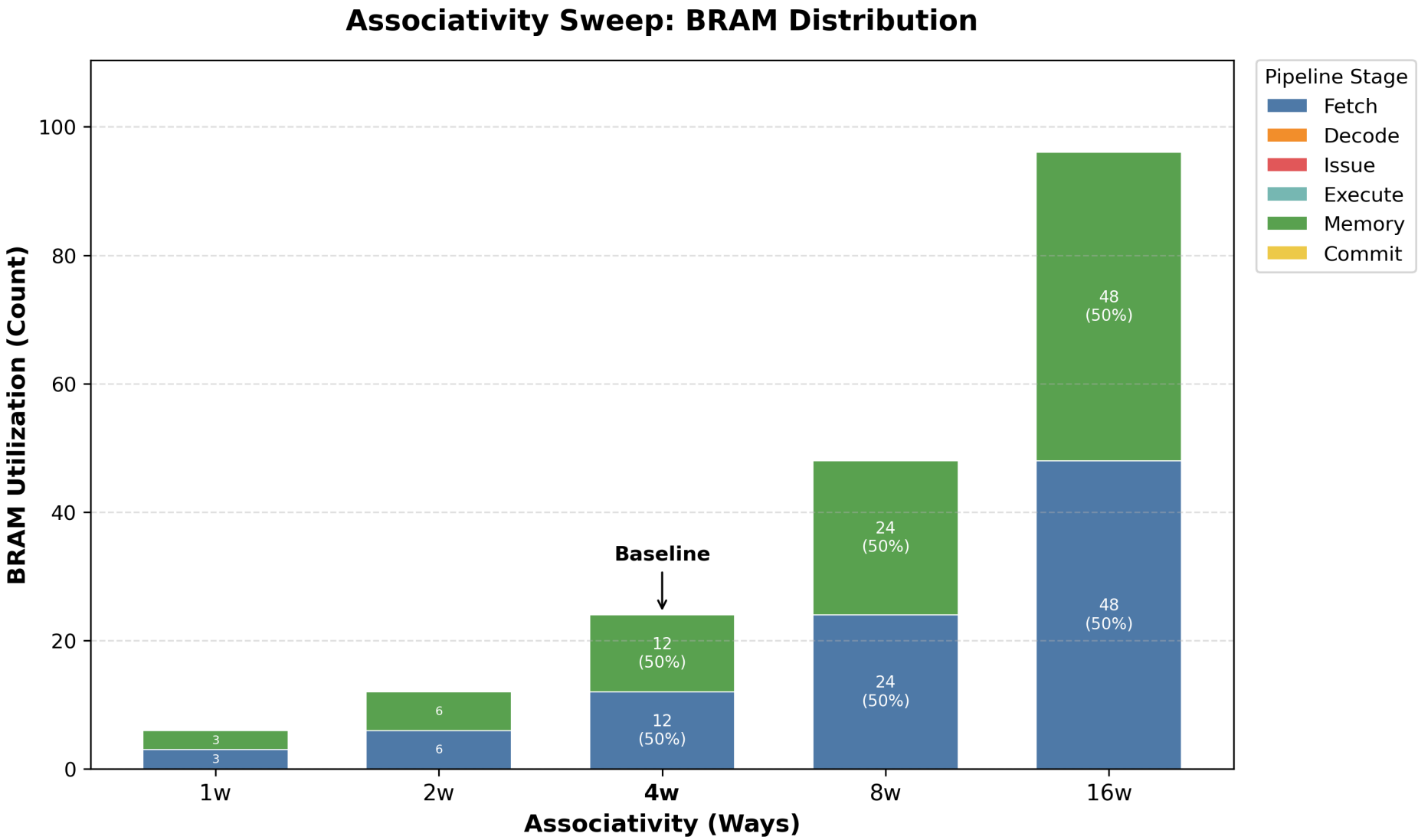


Resource Usage Dynamics



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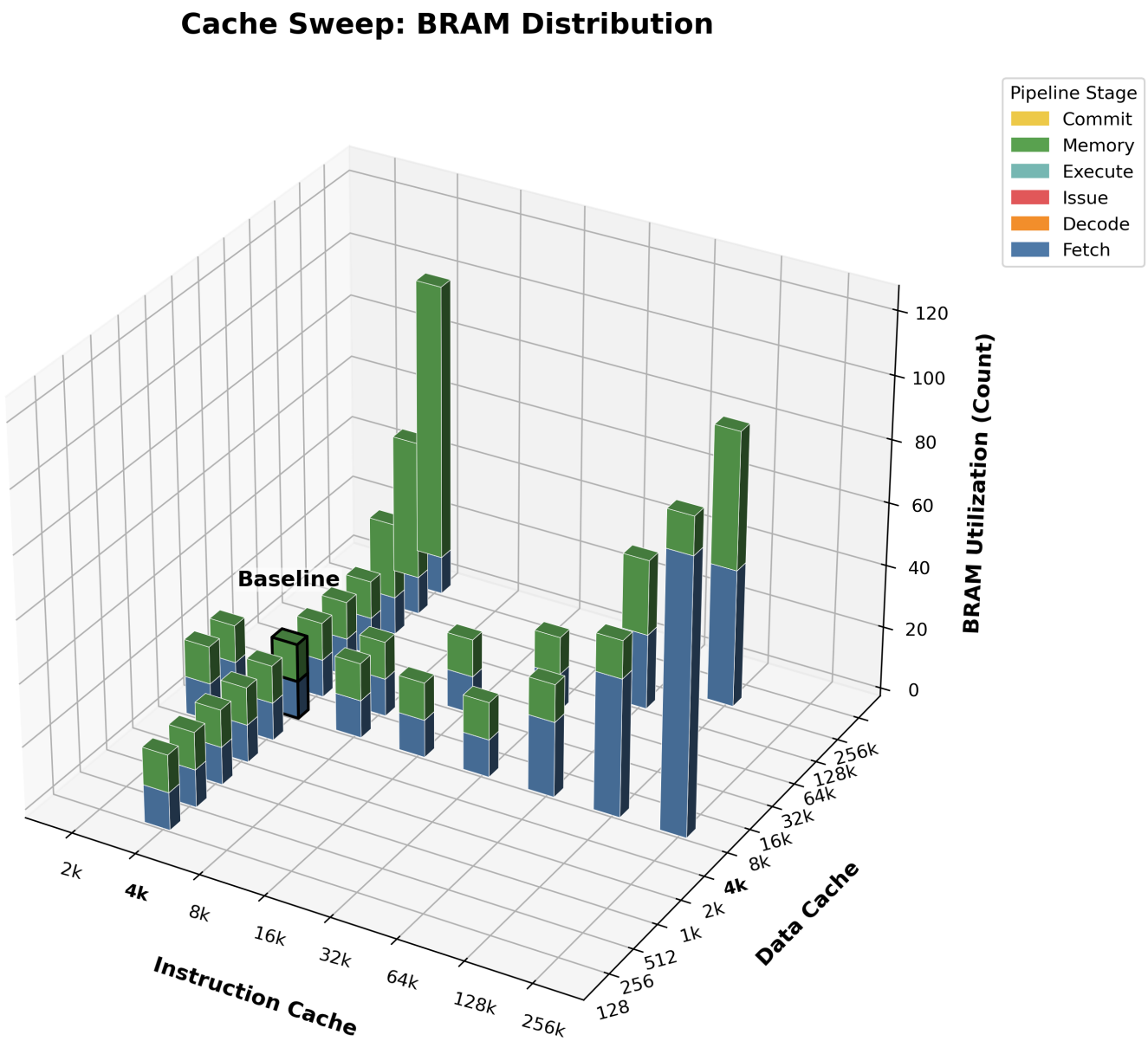

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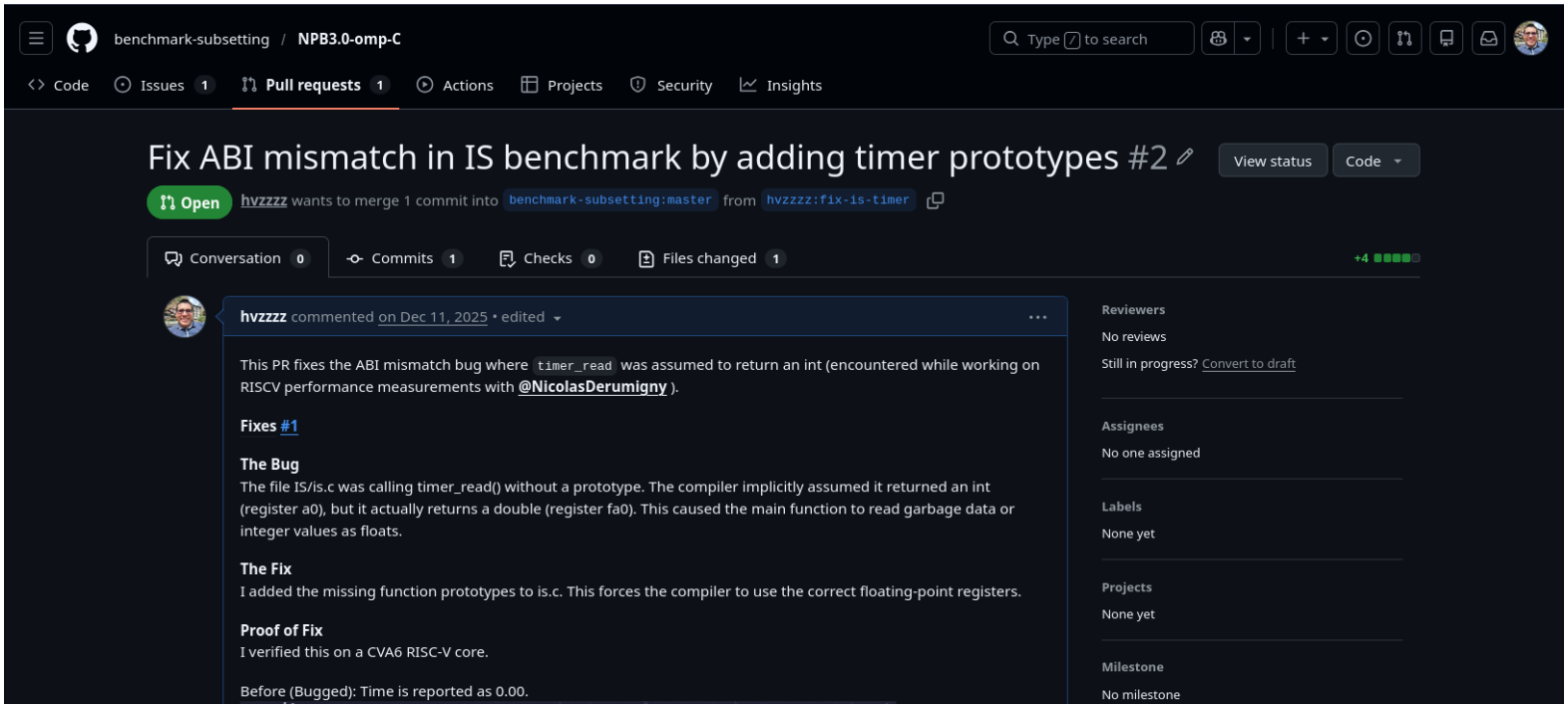
Contributions

- PR to the NPB repository

https://github.com/benchmark-subsetting/NPB3.0-omp-C/pull/2



- Automated tool for benchmarking architectural changes on the PYNQ Z2
- Baseline performance / utilization measurements for CVA6 multi-core

Outline

Conclusion

- *Automated Benchmarking Established*: Successfully developed an automated evaluation tool for the PYNQ-Z2 , securing complete single-core performance and resource baselines.
- *Cache Size Trade-offs*: Larger instruction and data caches significantly accelerate workloads like context switching and system calls , though they require substantially more BRAM.
- *Inefficient Scaling Returns*: Scaling cache associativity and branch predictors (BTB/BHT) yields negligible performance improvements while heavily increasing LUT and BRAM consumption.
- *Prepared for Multi-Core Expansion*: The established baselines clear the path for the next project phase: developing the shared resource arbitrator and decoupling the core.

Future Work

Stage 4 (Core Modification & Decoupling)	**Stage 5 (Arbitration Module Design)**	**Stage 6 (System Integration & Evaluation)**
Development of a modified CVA6 version	Development of the shared resource arbitrator	Evaluation of performance and FPGA utilization (FF/LUT/BRAM)
Implementation of an external module interface [optional]	Implementation of pipeline stalling logic [optional]	Evaluation of performance NPB and Unixbench
Enabling connection for FPU management [optional]	Routing of results between multiple cores [optional]	

Thank you!

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This work is being supported by the **BENAGIL TEAM**